

Miranda Zheng

U.S. Citizen | (929) 855-4668 | miranda.mzheng@gmail.com | [linkedin.com/in/miranda-mzheng](https://www.linkedin.com/in/miranda-mzheng) | github.com/mirmirmirr

EDUCATION

Rensselaer Polytechnic Institute | **GPA: 3.90**

Troy, NY

B.S. Dual, Computer Science & Information Technology and Web Sciences

May 2026

Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Web Development, Computer Organization, Programming Languages, Managing IT Resources

Activities and Leadership: Amateur Photographer, Teaching Assistant, Resident Assistant, HackRPI Director of Marketing, Chinese American Student Association Graphics Chair

EXPERIENCE

Lockheed Martin

May 2025 - Present

Software Engineer Intern

- Designing and implementing GitLab CI/CD pipelines to automate building, testing, and deployment of C/C++ applications and RPM packages in Red Hat Linux (RHEL) environments.
- Building Docker containers for RHEL, managing base images, dependencies, and deployment for product testing and release.

Johnson & Johnson

September 2024 - December 2024

Data Analyst

- Cleaned and standardized data from 18 Johnson & Johnson manufacturing sites to ensure GMP compliance.
- Conducted Product Quality Review (PQR) analyses on manufacturing data, identifying key reporting gaps and discrepancies, improving data accuracy by 44%.

Lockheed Martin

May 2024 - August 2024

Software Engineer Intern

- Refactored a customer-facing interface using Java Swing in Apache NetBeans, contributing to 2 software releases that improved functionality and user accessibility, while managing version control with TortoiseCVS.
- Optimized Java code by implementing inheritance-based class structures, reducing code redundancy and improving maintainability, leading to 30% faster development cycles.

NASA & Rensselaer Polytechnic Institute

January 2024 - May 2024

Student Researcher

- Translated equations into MATLAB, improving accuracy and speed of temperature and vaporization calculations, collaborating closely with research mentors.
- Researched and applied image enhancement techniques using Python and Mask R-CNN to analyze NASA ISS footage, improving visualization of experiment footage.

PROJECTS

Asian Cafe Website & POS System | *Next.js, Django, PostgreSQL*

December 2024 – Present

- Developed a Next.js-based web platform, optimizing SEO and enhancing customer experience for a restaurant, resulting in a streamlined ordering process and improved operational efficiency.
- Built a Django backend following the MVC architecture, with a PostgreSQL database managing 100+ menu items and 1000+ relationships to support complex order processing.
- Designed a RESTful API in Django to connect the frontend and Flutter-based POS system, ensuring data flow.

FlyBy | *React, Python, FastAPI, PyTorch*

March 2025

- Achieved second place in the 2025 NSBE x SHPE Climate Change & Sustainability Hackathon.
- Developed a full-stack web application utilizing FastAPI and React to detect and analyze marine debris in aerial footage using a YOLOv5 deep learning model trained on the TACO dataset.

Cool Academic RPI Bot (CARPI) | *Python, FastAPI, MySQL, React*

January 2023 – May 2025

- Developed CARPI Bot, a Discord bot for academic support and student engagement, in collaboration with an open-source team using Agile, implemented features like text-based calculators and academic calendar access.
- Led a team of 5 in designing and developing the frontend for a four-year course planner website, optimizing the user experience for both mobile and desktop users.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, HTML/CSS, PHP, SQL (Postgres/MySQL), NoSQL

Frameworks/Libraries: React, React-Native, Next.js, RESTful APIs, Django, FastAPI, MongoDB

Tools: Git, Github, TortoiseCVS, Figma, Microsoft Azure, Excel, Agile, Scrum